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Measuring the Speed of Earth Around the Sun Using a Classroom-Built Radio Horn Telescope

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Even though many physics teachers take their students on a calculation adventure through circular motion and Newton's universal law of gravity to determine Earth's velocity, most of us leave it at that. We present the final result and say, "Look, Earth is moving around the Sun at about 107,000 km/hr (66,000 mph), yet we can't feel the motion at all!" Well, thanks to an NSF-funded Research Experience for Teachers,[1] some eager participants, a simply built radio horn telescope pointed at the center of the Milky Way, and the addition of Doppler effect considerations, we can verify our calculations through observation... science!

The model

There is great versatility in creating lessons about Earth's orbit. In a general astronomy course, we can talk about Earth orbiting the Sun in a nearly circular orbit. This includes discussing the scale of astronomical objects and distances. This can aid in introducing scientific notation, which leads to defining the average distance from Earth to the Sun as a unit, the astronomical unit (au) = 1.496×10^{11} m. Most students know that Earth takes one year to complete one revolution around the Sun and can convert that time into seconds. Without considering the subtle changes for Earth's elliptical orbit, they can run through a circular motion argument to get

$$v = \frac{2\pi r}{T} = \frac{2\pi(1\,\text{au})}{1(\text{year})} = 29.8\frac{\text{km}}{\text{s}}. \tag{1}$$

In our physics classes, we can look at Newton's universal law of gravity and centripetal force to calculate the velocity.

$$\frac{GM_s m_e}{r^2} = \frac{m_e v^2}{r}. \tag{2}$$

G = universal gravitational constant
M_s = mass of the Sun
m_e = mass of Earth
r = distance between Earth and Sun
v = velocity of Earth

Since the mass of the Sun is derived from our speed around the Sun, when the students search out and use that mass of the Sun, Eq. (2) always works out very nicely…29.8 km/s! However, once the answer is obtained, there is a good chance this is where most of us finish the lesson, with no easy classroom method to verify with observation.

Radio horn telescope to the rescue!

What does a radio telescope have to do with the speed of Earth? Before answering that question, we first have to realize that the Milky Way is full of hydrogen. Some of the hydrogen is locked up in the hot stars we see all around us, but a lot of it is in huge clouds of cold neutral hydrogen that pervades the whole galaxy. It turns out that even this cold hydrogen emits radiation as a result of a hyperfine transition of the electron called a spin-flip transition. It is a low-energy radio emission with a very well-known frequency of 1420.406 MHz (also commonly known by its wavelength, the 21-cm line). This radio emission is the bread and butter of research for many radio astronomers who are looking at rotation curves of galaxies. The curves use Doppler to plot velocities of galactic material as a function of distance from the galaxy's center. These curves show higher than expected speeds of outer galactic material, and have led to the idea of dark matter.

Fig. 1. An array of four horn scopes with the 100-m Green Bank Telescope in the background. Green Bank, WV.

The small radio horn telescope (Figs. 1, 5, and 7) finally enters the picture when a band of like-minded high school science teachers became involved with Digital Signal Processing in Radio Astronomy (DSPIRA). DSPIRA was a three-year NSF-funded Research Experience for Teachers[2] summer project, run through West Virginia University from 2017-2019. (A 2020 summer follow-on effort led to the creation of a STEM-curriculum website for all teachers to access many of the DSPIRA activities,[3] including instructions for building the horn, and a lesson sequence to duplicate this experiment.) These classroom-built horn radio telescopes (< $200) have proven to be very good at detecting neutral hydrogen within the Milky Way.

In July 2018, while analyzing some of the data of the galactic center during our DSPIRA work, I was bothered that the

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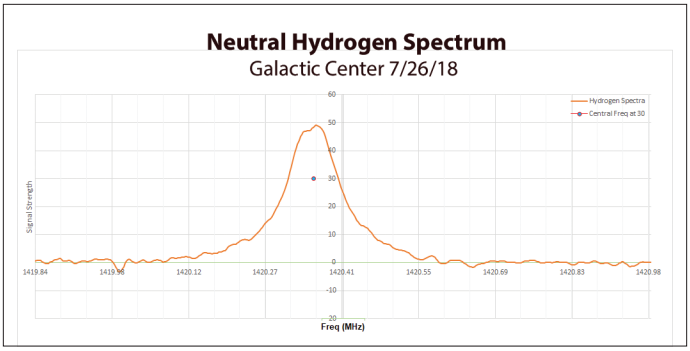

Fig. 2. My original spectrum of the galactic center. The peak is a lower frequency (red-shifted) from the neutral hydrogen frequency of 1420.41 MHz.

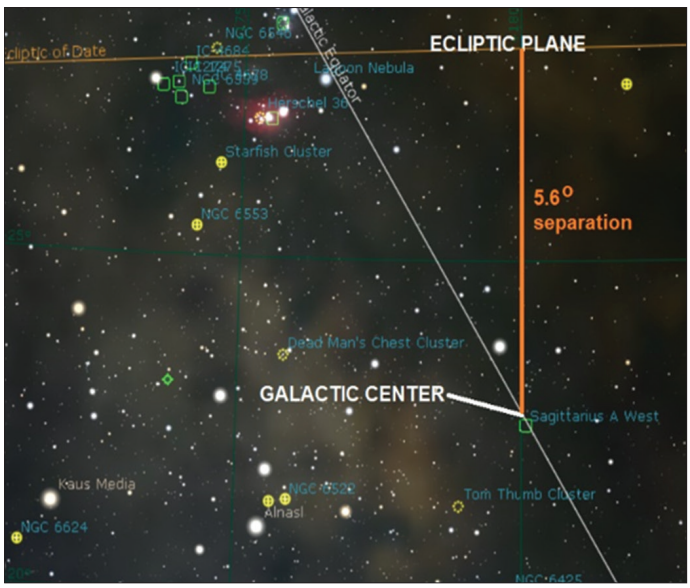

Fig. 3. Stellarium software[4] look at galactic center and ecliptic plane.

center of the peak of the hydrogen spectrum was skewed from the 1420.406 MHz value (Fig. 2). My initial reasoning was that if the Sun is going around the Milky Way in a circle, then there should be no radial motion towards the center, just a random distribution of speeds centered around zero. However, there was a non-zero radial component. I questioned whether Earth's 30 km/s was enough so that a Doppler effect would be detectable, and it turned out to be very feasible. I decided to take on this idea and turn it into a year-long experiment looking at the center of the galaxy. The whole idea does lead to a powerful question and assumption. In a large spiraling galaxy moving through the universe, what is the most fixed reference point to look at within the galaxy? I decided that our galactic center would be the best bet.

Further development of the model

Since the Doppler effect relies on the radial velocity (moving towards or away from) component of the overall velocity, there is some need for trigonometry. It is also important to know how much of Earth's motion around the Sun is oriented towards the galactic center. Fortunately, the plane of Earth's orbit, known as the ecliptic plane (see Fig. 3), comes within 5.6° of the galactic center. Why "fortunately"? Imagine

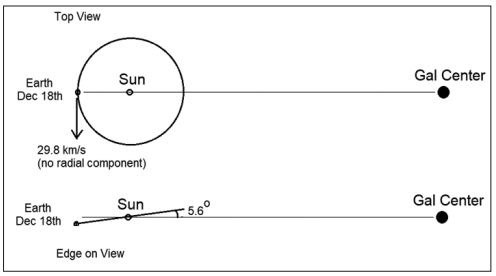

Fig. 4. Top and edge on view of Earth's orbit.

looking at a white dot painted onto a bicycle tire. If the tire is rotated while you are looking face on, you would see the white dot moving in a circle, but never getting closer or farther away from you…thus no Doppler. Edge on would be the ideal situation, with the white dot moving cyclically towards and away from you. Our situation is very near ideal, as illustrated in Fig. 4. It will involve a small modification to the model equation.

$$v_{\text{radial max}} = v \cos(5.6°) = 29.6 \text{ km/s}. \quad (3)$$

It turns out that Dec. 18th (±1 day) is the day in the calendar year that the Sun aligns closest to the galactic center in our sky. This is analogous to the Earth, Moon, and Sun alignment for a new moon. So, Dec. 18th will become Day 0 in the model (Fig. 4).

Another simplification of the model will be to assume Earth follows a circular path. It is actually elliptical with speed variations from closest to Sun (perihelion – Jan. 3) speed at 30.3 km/s and farthest from Sun (aphelion – July 3) at 29.3 km/s. The ±0.5 km/s will be ignored in this consideration.

This all leads to a model equation for the radial component of Earth's motion around the Sun as

$$v(N) = v_{\max} \sin\left[\left(\frac{360°}{365.25 \text{ days}}\right)N\right] \quad N = \text{Day \# from Dec.18th} \quad (4)$$

This turns into

$$v(N) = 29.6 \text{ km/s} * \sin(0.986N). \quad (5)$$

This equation will generate the model graph to which the actual data points will be compared.

The Doppler effect – A quick review

The Doppler effect is in my state's (NY) physics curriculum. It is only qualitative: red shift: moving away, frequency lower, wavelength longer. However, the low relative speed of Earth's motion compared to the speed of light allows the use of the non-relativistic Doppler equation, and thus the equation turns into a simple ratio for calculating speeds compared to c, the speed of light.

$$\frac{f' - f_0}{f_0} = \frac{v}{c}. \quad (6)$$

f' = frequency determined from spectrum
f_0 = neutral hydrogen frequency 1420.406 MHz
c = speed of light, 3.0 x 10^5 km/s
v = measured radial speed of Earth with respect to galactic center (+ is blue shifted)

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Fig. 5. The DSPIRA Horn Scope was designed after the original horn to detect neutral hydrogen in the Milky Way in 1951, which is currently on display at the Green Bank Observatory facility.

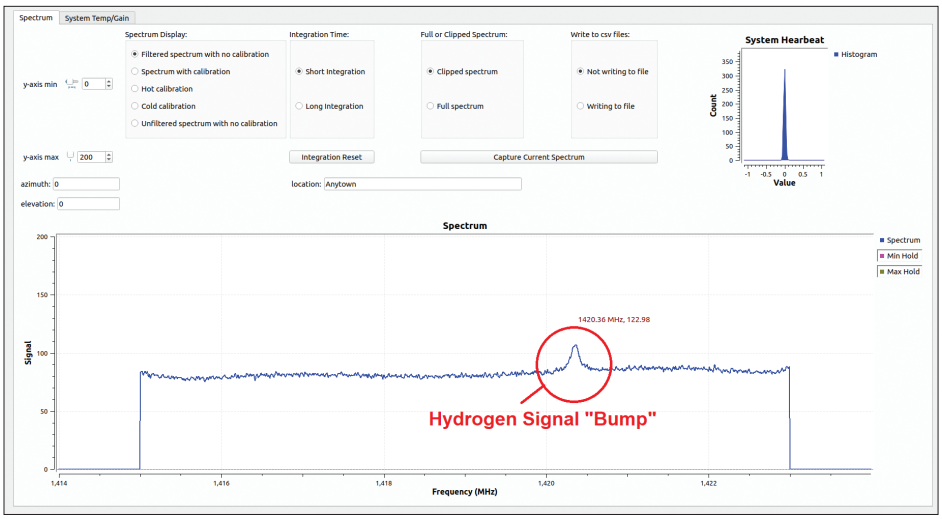

Fig. 6. A view of the GNU radio spectrum with hydrogen.

Fig. 7. Aryan, Jared, and Gianna staying after school to take two readings of the galactic center crossing the meridian in October.

Radio astronomy, the horn, and GNU Radio

The typical image of a high school astronomy observation session is a nighttime scene with optical telescopes. However, since the atmosphere is completely transparent to the radio waves in the range of neutral hydrogen, radio astronomy can be done anytime. A great teaching benefit of this fact is that observations can be made during daylight hours when the students are a captive audience. The spectrometer at the back end is run with the open-source software GNU Radio. The DSPIRA horn scope is similar in design to that used by Ewen and Purcell[5] of Harvard University. They used the horn in Fig. 5 to make the first detection of 21-cm neutral hydrogen back in 1951.

A typical galactic center spectrum is seen in Fig. 6. It is a digitized signal that has a frequency resolution of 2.44 kHz. This converts to a Doppler shift speed resolution of about 0.5 km/s, which is well within Earth's ± 30 km/s being measured.

Again, for classroom considerations, the dataset of frequency vs. intensity can easily be exported from the GNU Radio-DSPIRA spectrometer into familiar spreadsheet programs for student analysis.

The measurements

In order to get the best consistent measurements, the spectra are taken as the galactic center is crossing the local sky's meridian. This allows the galactic center to be at its highest point above the horizon, which from my location in New York is only 19° above the horizon. Also, when viewing something at the meridian, Earth's rotational speed has no radial component towards the object. The "fun" part of making these meridian observations is that over the course of a year, the observation time drifts from taking measurements in convenient October afternoons, through to getting up at 3:00 a.m. in May to take a reading. From a classroom teacher's perspective, the time of meridian crossing is convenient from October when it occurs in the late afternoon (Fig. 7) through to March when the crossing is

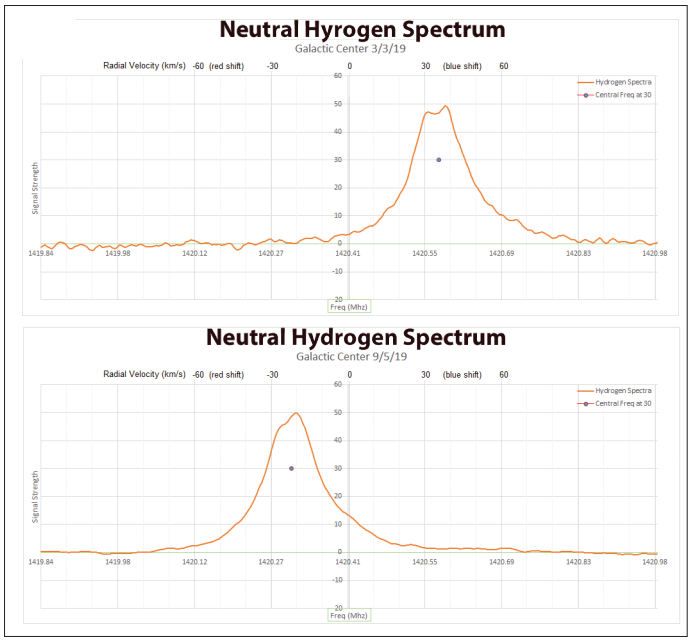

Fig. 8. Two spectra at a separation of about six months, and near the maximum radial speeds, towards and away. It clearly shows the shift over time. The radial velocity scale is on the top of graph. The grid spacing is 10 km/s.

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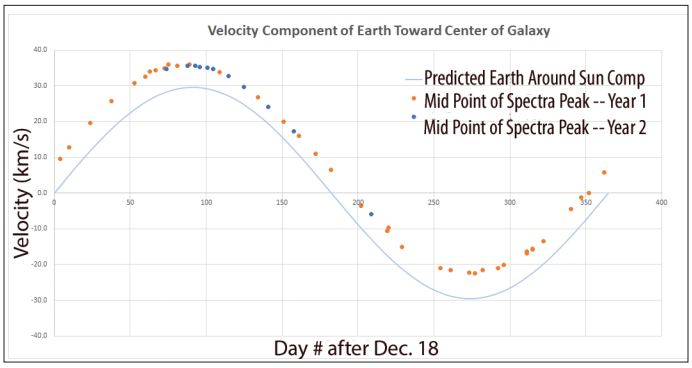

Fig. 9. The final year-long+ dataset of actual recorded radial speeds vs. the model graph.

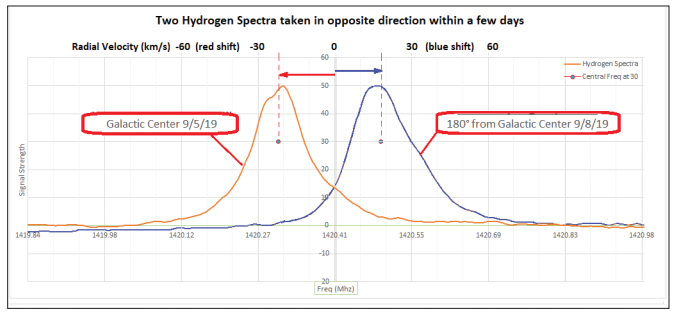

Fig. 10. Two spectra taken almost the same day, but in opposite directions. The velocities show the close "equal and opposite" nature expected, at speeds of nearly 20 km/s.

early morning, just before school starts. This would mean that even just two viewings, at six-month separation, would yield some clear classroom results (Fig. 8).

The results

When looking at the neutral hydrogen "bump" in the spectrum, a singular frequency needs to be determined to represent the average for use in the equations to determine a radial speed. To determine "the" frequency to use in the calculation for speed, the midpoint at the 3/5-max, or vertical 30 (arb units) mark, was found. For students, this is a good chance to work either with interpreting graphical data manually, or to learn to work with spreadsheet data. In either case, it is visually easy to observe the significant frequency shift over the months (Fig. 8).

The center observed frequency is run through Eq. (6) and plotted against the model graph generated from Eq. (5). Figure 9 is the resulting graph of radial Earth speed vs. day# after 15 months of collecting data. The first thing observed is the plot points are depicting the very clear sine wave nature of the radial component of the velocity expected. It has been very rewarding to see that the data running into the second year are laying over the first year's data nicely. The next question that comes up in looking at the data is, "Why are the data skewed from the model?" One of the assumptions of the model was that the Sun was in a circular motion around the galaxy. It turns out that it is not. Though, still a tough number for astronomers to nail down, published values for the radial component of the Sun's velocity, with respect to the galactic center, range from about 8 km/s to 14 km/s towards the galactic center.[6-8] So, as an unexpected bonus, this little radio horn's data are clean enough to test the current value of the Sun's radial velocity within the galactic plane. This leads to the need to modify Eq. (5) with an additional term to account for the vertical skew in the data.

$$v(N) = v_{\text{radial Sun}} + 29.6 \text{ km/s}^{*}\sin(0.986N) \qquad (7)$$

Since there are now over 50 data points, the radial component of the Sun's velocity from the horn data can be evaluated against the model's predicted value. This was done simply by taking the difference between each data point's calculated speed and the predicted speed for that day. Then, the collective differences were run through basic statistics to determine mean and standard deviation. This leads to a determination of the Sun's radial component of velocity towards the galactic center to be 7.0 ± 1.0 km/s. This brings it into shouting range of the currently acceptable value of about 10 km/s. This is very respectable for the little horn scope.

One of my DSPIRA colleagues suggested that I should look the "other way," or at the 180° point from the center of the galaxy. We galactic travelers should be "passing" things in the outer reaches of the galaxy that are at 180° from the center. Therefore, using the same argument as with looking at the galactic center, there should be no Doppler shift, except Earth's motion and the Sun's radial velocity towards the galactic center. So, a new dataset was started by looking 180° away from the galactic center. If you are moving towards the center, then you should be moving away from the opposite point. Therefore, the velocities should be equal and opposite. The two spectra in Fig. 10 were taken on nearly the same day. They indicate the expected "equal and opposite" nature reasonably well, both being close to 20 km/s towards the center and 20 km/s away from the opposite point from the center.

The overall graph for looking at 180° from the center (Fig. 11) shows a relatively good fit of the model curve, and it does show the opposite skew as expected. The graph is a little "rougher" looking than the galactic center graph (Fig. 9). The method of pointing the scope at the needed 78° elevation for these spectra proved to be more susceptible to being off, possibly leading to the higher deviation seen. Greater care in

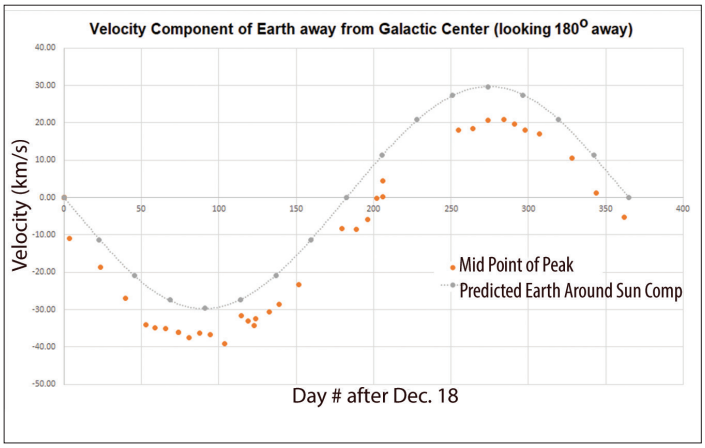

Fig. 11. The year-to-date data for looking outward in the galaxy at 180° from center. The skew of data, this time, is below the model, as would be expected.

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pointing in the future should reduce this error. Still, the radial component of the Sun's velocity is –9.1 ± 1.7 km/s, which is still in good agreement with the galactic center observations.

DSPIRA legacy

A tremendous thanks needs to go to the principal investigators at West Virginia University for developing this program. This paper is the result of one investigation that could be done with the scopes. NSF's and the DSPIRA group's hopes is that other high school and college teachers will find this interesting enough to take on this project and add an observational tool to explore aspects of the Milky Way. The scope itself is a simple design that could be built by students in a construction class or for after school projects. The horn and can assembly is under two pounds, and our stand design made of 2x4s is quick to assemble and break down for easy storage. My stand for 80%+ of my data was my wheelbarrow. I would roll it, the horn, and laptop out of my garage and into the yard. I would sit the horn, pointing south, in the tray bed of the wheelbarrow, then use a leveling app on the phone to align the altitude angle. For my 3:00 a.m. readings, I could wake up, set up, get my data recorded, and be back to bed in half an hour. My small student groups (Fig. 7) needed about a 15-minute tutorial for using the scope to get data. Using the DSPIRA website[3] would be crucial for getting started from scratch and finding information on getting the correct electronics and current software.

Summary

The creation of this low-cost radio telescope has opened up an avenue for new experiments within the reach of high school teachers and students. With this instrument, I was able to take a selection of topics that we all cover in a general introductory physics course and turn it into a unique experiment that carries the students well outside the classroom, and not far from where modern-day astronomers are currently exploring aspects of the Milky Way galaxy.

Galileo supposedly muttered, "And yet it moves!" after he retracted his claim of a heliocentric universe (to avoid the punishment for heresy in the eyes of the Church). Well, here was an actual experiment within reach of the classroom to verify that fact. The resulting data from the 15-month experiment demonstrated the expected nature of the radial speed towards the center of the Milky Way galaxy in both its sinusoidal shape and magnitude. As the data were collected, there was a consistent skew to the model that led to the realization that this experiment was also measuring the radial component of the Sun's motion around the center of the galaxy. With experimental results of 7.0 ± 1.0 km/s from looking towards the center, and 9.1 ± 1.7 km/s from looking in the complete opposite direction, there is recognizable agreement with the current, but still relatively uncertain, values from the professional astronomers that are about 8 to 14 km/s. It was a great reward for me, as a 30-year physics teacher, to actually perform a high school experiment that could verify that Earth does indeed move around the Sun!

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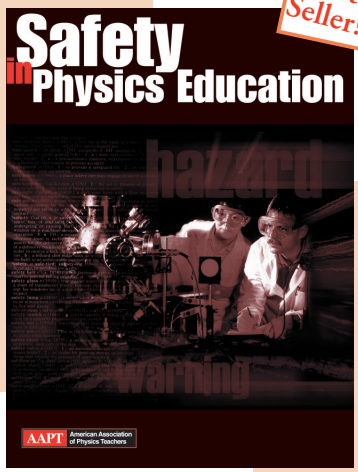

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